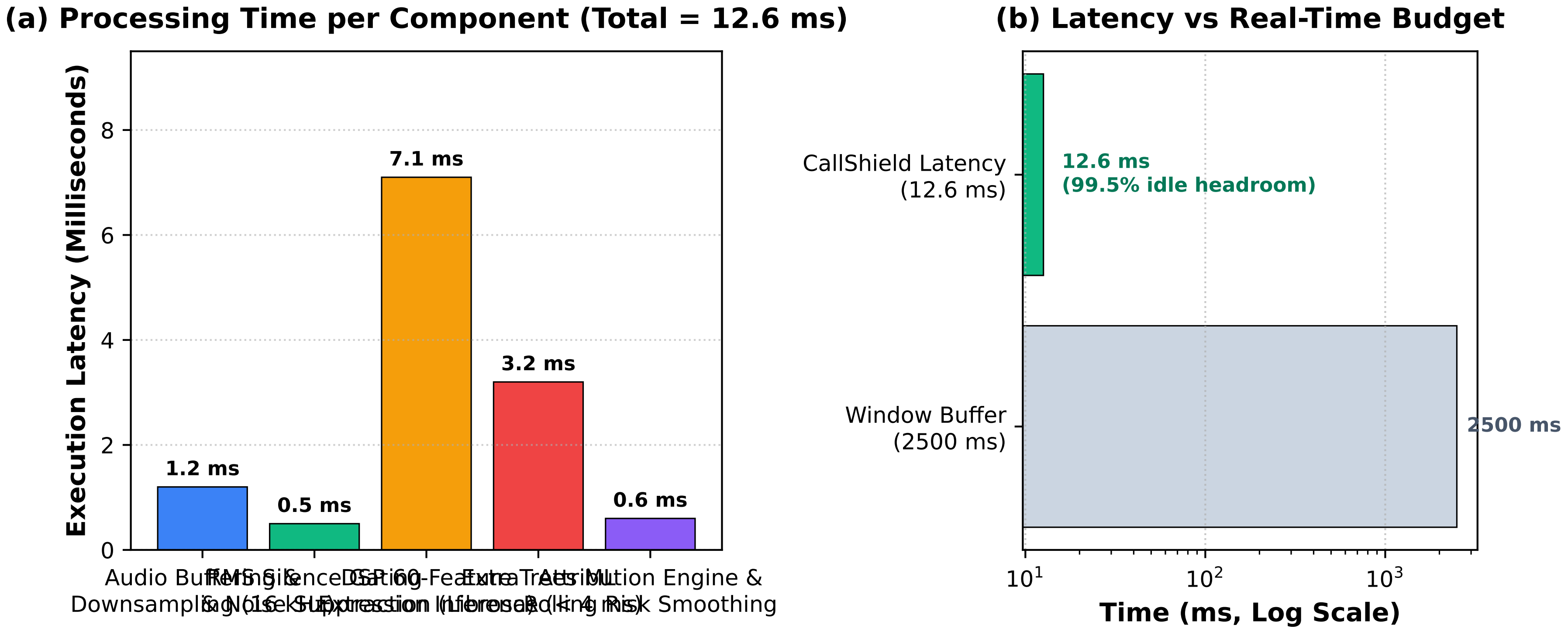
(a) Processing Time per Component (Total = 12.6 ms)
Execution Latency (Milliseconds)
0
2
4
6
8
1.2 ms
0.5 ms
7.1 ms
3.2 ms
0.6 ms
Audio Buffering &
Downsampling (16 kHz)
RMS Silence Gating
& Noise Suppression
DSP 60-Feature
Extraction (Librosa)
Extra Trees ML
Inference (< 4 ms)
Attribution Engine &
Rolling Risk Smoothing
(b) Latency vs Real-Time Budget
CallShield Latency
(12.6 ms)
12.6 ms
(99.5% idle headroom)
Window Buffer
(2500 ms)
2500 ms
10^1
10^2
10^3
Time (ms, Log Scale)